Lecture 8

↳ continue multi-qubit states, gates, and measurement.

↳ $|\psi\rangle = |\Lambda\rangle \otimes |\Phi\rangle$

↳ $C = A \otimes B$

↳ $\underbrace{C\psi = (A \otimes B)(|\Lambda\rangle \otimes |\Phi\rangle) = (A|\Lambda\rangle) \otimes (B|\Phi\rangle)}_{\text{convince by example that this is true.}}$

Does it make sense?

$$C = \underset{q_1}{I} \otimes \underset{q_0}{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$|\psi\rangle = |\Lambda\rangle \otimes |\Phi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$C|\psi\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \overset{?}{=} |0\rangle \otimes |0\rangle \rightarrow |01\rangle$$

$$I|\Lambda\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X|\Phi\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$(I|\Lambda\rangle) \otimes (X|\Phi\rangle) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |0\rangle \otimes |0\rangle$$

makes sense

Prof did the example of $C = \underset{q_1}{I} \otimes \underset{q_0}{Z}$

$$Z_0 = \underset{q_1}{I} \otimes \underset{q_0}{Z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$Z_1 = Z \otimes I = \begin{pmatrix} 1[I] & 0[I] \\ 0[I] & -1[I] \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$Z_0 |10\rangle \overset{?}{=} (+1)|10\rangle$$

$$\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = (+1)|10\rangle$$

$$Z_0 |01\rangle = (-1)|01\rangle$$

$$\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = (-1) \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\underbrace{(Z \otimes X)}_{\text{get the order right!}}(|1\rangle \otimes |0\rangle) \overset{?}{=} (Z|1\rangle) \otimes (X|0\rangle) = (-|1\rangle) \otimes |1\rangle$$

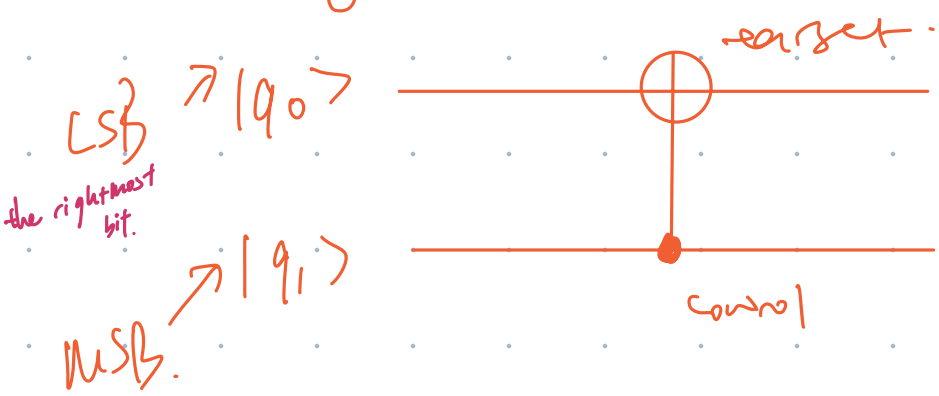
$$\begin{bmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \\ 0 & \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

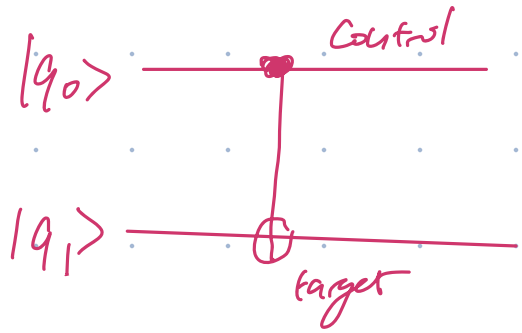
$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$(Z \otimes X)(|1\rangle \otimes |0\rangle) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = -\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = -|1\rangle \otimes |1\rangle$$

CNOT Gate (aka CX)

↳ specify which is the control / which is the target





choose this one for the example

Derive 4x4 rep of CNOT from Dirac Notation for the truth table.

target

control

$$\text{CNOT}\,|00\rangle = |00\rangle$$

$$\text{CNOT}\,|01\rangle = |11\rangle$$

$$\text{CNOT}\,|10\rangle = |10\rangle$$

$$\text{CNOT}\,|11\rangle = |01\rangle$$

wow, cool!

$$\text{CNOT} = |00\rangle\langle 00| + |11\rangle\langle 01| + |10\rangle\langle 10| + |01\rangle\langle 11|$$

analogy

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|$$

$$= \begin{pmatrix}1\\0\end{pmatrix}(0\ 1) + \begin{pmatrix}0\\1\end{pmatrix}(1\ 0) = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$$

$$|00\rangle\langle 00| = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}(1\ 0\ 0\ 0)^* = \begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

$$|11\rangle\langle 01| = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}(0\ 1\ 0\ 0)^* = \begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&1&0&0\end{pmatrix}$$

$$|10\rangle\langle 10| = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}(0\ 0\ 1\ 0)^* = \begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&1&0\\0&0&0&0\end{pmatrix}$$

$$|01\rangle\langle 11| = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}(0\ 0\ 0\ 1)^* = \begin{pmatrix}0&0&0&0\\0&0&0&1\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

CNOT = [ ] + [ ] + [ ] + [ ]

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Construction in Pauli Matrix: ?

$$CNOT = I \otimes \left(\frac{I+Z}{2}\right) + X \otimes \left(\frac{I-Z}{2}\right)$$

$$C = \frac{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}{2} = \frac{\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}}{2} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathrm{I}.$$

$$\frac{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}{2} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad \mathrm{II}.$$

$$\mathrm{I} + \mathrm{II} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \Rightarrow \text{verified.}$$

What happens if control is in a superposition of $|0\rangle$ and $|1\rangle$?

$$|\psi\rangle = (I \otimes |+\rangle)(|0\rangle \otimes |0\rangle) = |0\rangle \otimes |+\rangle$$

$$= |0\rangle \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle\right)$$

How to understand entanglement.

$$CNOT|\psi\rangle = \frac{1}{\sqrt{2}} CNOT\left(|00\rangle + |01\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\underbrace{|00\rangle + |11\rangle}\right)$$

entangled state

?

how do you know that it's entangled

$\neq |\psi_1\rangle \otimes |\psi_0\rangle$

not a product state of any 2 basis states.

$$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} = |++\rangle$$

this is a product state

how do we know that the 2 qubits have the same state?

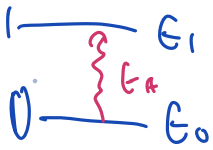
$$\underbrace{|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle}$$

A & B's qubit can either be both $|0\rangle$ or both $|1\rangle$ when measured in the Z-basis

↳ but when you express this in the X-basis

↳ how do you detect entanglement.

How do you actually do a CNOT gate?



→ using lasers to excite atoms/electrons/whatever

↳ equivalent to doing rotations on the bloch sphere

$E_A = E_1 - E_0$

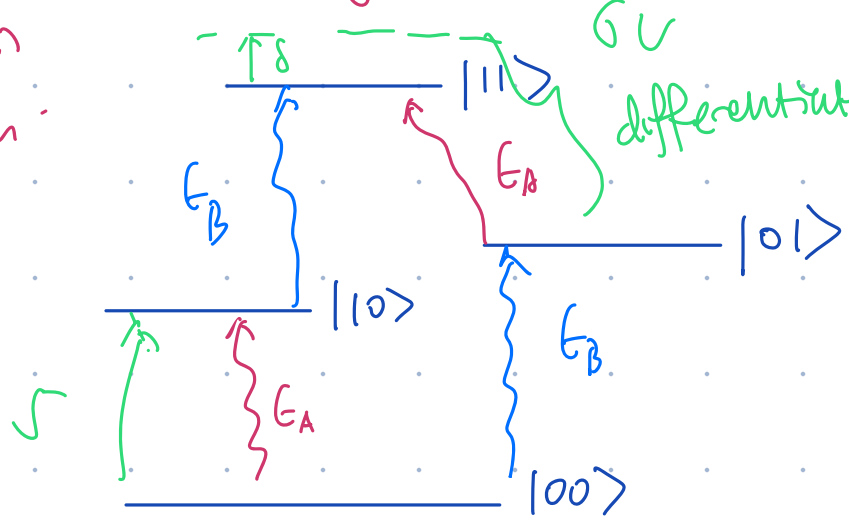
$$H = \underbrace{E_A |1\rangle\langle 1| \otimes I}_{\text{when qubit 1 is excited. Energy} = E_A} + \underbrace{E_B\, I \otimes |1\rangle\langle 1|}_{\text{when qubit 0 is excited Energy} = E_B} + \delta |11\rangle\langle 11|$$

both excited:

energy $= E_A + E_B$.

Suppose

haha this is hilarious.



differentiate between the 2 energy levels.